

DESIGN PROJECT - II REPORT

On

CROP DISEASE PREDICTION USING EXPLAINABLE AI

Submitted in Partial Fulfillment of Award of
BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

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CERTIFICATE

This is to certify that the Design project – II work entitled “Crop Disease Prediction Using Explainable AI”, submitted by K. Nithin Kumar[2022BCSE07AED659], R.J.Sumathi[2022BCSE07AED661] ,N.VenkataSubbareddy[2022BCSE07AED658], P.LakshmiManojKumar[2022BCSE07AED660],andD.DeviVaraPrasad[2022BCSE07AED662] in partial fulfillment for the award of the degree of Bachelor of Technology (Department of CSE) of Alliance University, is a bonafide work accomplished under our supervision and guidance during the academic year 2024-2025. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

I/We hereby declare that the Design project - II entitled **“Crop Disease Prediction Using Explainable AI”** submitted by us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE) of Alliance University, is a record of our work carried under the supervision and guidance of Dr. Mir Wajahat Hussain

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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PREFACE

Agriculture is vital for global food supply and livelihoods, but plant diseases pose a major challenge, leading to crop losses and financial hardships. Early and accurate disease detection is crucial for effective intervention and improved yields. Advancements in artificial intelligence (AI), particularly Convolutional Neural Networks (CNNs), have enabled the development of automated plant disease classification systems. This project leverages CNNs to identify and diagnose plant diseases from leaf images, helping farmers make informed decisions for better crop management. The project also incorporates Local Interpretable Model-agnostic Explanations (LIME) to enhance model explainability, ensuring transparency and trust in AI-driven solutions. The CNN model is trained on a comprehensive dataset of plant images, and its performance is evaluated using metrics like accuracy and confusion matrices. While promising results were achieved, challenges such as dataset diversity, image quality, and resource-intensive training remain. This document provides an overview of the project's methodology, implementation, results, and potential for future enhancement in agricultural practices.

ABSTRACT

Plant diseases can have profound effects on the economy, impacting both local and global scales. These diseases can lead to substantial losses in agricultural productivity, affecting crop yields and quality. In this context, deep learning algorithms are widely acknowledged as effective solutions. However, the use of these black-box approaches raises concerns about trust in interpreting and validating the decisions generated by the models. This study proposes an explainable artificial intelligence (XAI) based plant disease classification system to classify and identify distinct ailments with improved accuracy. The system correctly identifies 38 different plant diseases with accuracy, precision, and recall as 99.69%, 98.27%, and 98.26%, respectively. These predictions are subjected to additional analysis employing the local interpretable model-agnostic explanations (LIME) framework to produce visual explanations aligning with prior beliefs and adhering to established best practices in explanations. This system will serve as a promising avenue for revolutionizing disease detection, fostering informed decision-making, and ultimately contributing to global food security.

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LIST OF ABBREVIATIONS AND SYMBOLS USED

AI – Artificial Intelligence

XAI – Explainable AI

CNN – Convolutional Neural Network

LIME – Local Interpretable Model-Agnostic Explanations

AUC – Area Under the Curve

IPM – Integrated Pest Management

IoT – Internet of Things

ResNet – Residual Network

EfficientNet – Efficient Convolutional Neural Network (a type of model architecture)

DenseNet – Densely Connected Convolutional Network

Grad-CAM – Gradient-weighted Class Activation Mapping

SHAP – SHapley Additive exPlanations

F1-score – F1 Score, a measure of a model's accuracy

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CHAPTER 1

INTRODUCTION

Agriculture, the foundation of numerous economies and a vital provider of food security, faces constant challenges from different plant diseases that jeopardize both crop yields and quality. Crop diseases, usually triggered by fungi, bacteria, viruses, or environmental stressors, may result in substantial agricultural losses if they are not promptly identified and treated. Traditional methods for diagnosing plant diseases usually entail manual inspection. This approach can be time-consuming, expensive, and dependent on the knowledge of skilled agronomists

1.1 INTRODUCTION TO CROP DISEASE PREDICTION

Through the examination of plant images, machine learning models are able to comprehend subtle patterns linked to particular diseases, frequently achieving remarkable accuracy. Nonetheless, deep learning models are frequently perceived as "black boxes" due to their lack of transparency. The decision-making process within these models can be challenging to interpret. This limitation can impede user trust and diminish the effectiveness of AI-driven predictions for farmers and agricultural experts who depend on clear insights to make well-informed decisions. To tackle these challenges, the incorporation of Explainable AI (XAI) in predicting crop diseases has become increasingly popular. Explainable AI techniques empower models to not only forecast results but also offer valuable insights into the rationale behind specific predictions. By shedding light on the rationale behind a model's predictions, XAI closes the distance between intricate AI systems and user understanding, enhancing the reliability and usability of these models. This level of transparency enables agricultural experts to comprehend model predictions effectively, cultivating trust in AI suggestions and promoting their implementation in the field. In this essay, we delve into the significance of Explainable AI in predicting crop diseases. We specifically examine how XAI improves the interpretability of models, builds trust in AI systems, and facilitates better decision-making processes. We additionally explore the current status of XAI applications in agriculture and emphasize its capability to transform disease management practices, thereby enhancing sustainability and resilience in farming systems worldwide. Agriculture plays a vital role in ensuring food security and is a significant component of numerous economies. Nonetheless, it is constantly under threat from plant diseases. These diseases can result from fungi, bacteria, viruses, or adverse weather

conditions. They have the potential to cause substantial reductions in both crop yield and quality. If not promptly identified and controlled, these diseases have the potential to decrease food production, affecting the income of farmers and the supply of food. In the past, recognizing crop diseases has typically involved manual inspections, a process that can be time-consuming, expensive, and reliant on experts who may not be readily accessible, particularly in rural regions. Due to this, there is an increasing demand for quicker and more precise methods to detect crop diseases early. Artificial Intelligence (AI), particularly deep learning, has displayed encouraging outcomes in addressing intricate challenges in agriculture. This includes the prediction of crop diseases. When examining images of plants, AI models can identify patterns associated with particular diseases, frequently achieving high accuracy.

1.1.1 Introduction to disease detection

A significant obstacle associated with these AI models is their tendency to operate like "black boxes," making it difficult to comprehend their decision-making process. This lack of transparency can pose a challenge for farmers and agricultural experts to fully trust these models, as they might not comprehend the reasoning behind a specific prediction. To address this issue, the concept of Explainable AI (XAI) has been introduced. AI offers methods for AI models to elucidate their decisions clearly and comprehensibly. By incorporating XAI into crop disease prediction models, we facilitate comprehension for farmers and agricultural professionals, providing insights into the reasoning behind a model's prediction. This explanation enhances trust while boosting users' confidence in utilizing AI predictions for managing diseases. AI technology ensures that AI doesn't just provide answers, but also reveals the reasoning behind its responses. This added transparency enhances the reliability of AI systems for practical application. With the rising global population and shifting climatic conditions, challenges faced by farmers have intensified, particularly in identifying and managing plant diseases. These diseases, if left undetected, can significantly reduce crop yield and quality, leading to economic losses and endangering food security worldwide. Traditionally, disease diagnosis relies on manual inspection by experts, a process that is both time-intensive and subjective. Often, by the time visible symptoms emerge, the disease may have already spread, making control efforts less effective. Recent advancements in technology are transforming agriculture, providing innovative tools that support sustainable food production. In particular, machine learning (ML) techniques, such as image recognition, show promise in detecting plant diseases at early stages, allowing farmers to prevent spread and minimize crop damage. The early detection timeline varies across plant species and diseases, some requiring only a week, while others take longer for symptoms to appear. Numerous computer-aided diagnosis (CAD) systems have been developed to assist farmers in this regard, significantly improving diagnostic accuracy. By reducing reliance on manual inspections, CAD systems empower farmers with timely insights, ultimately enhancing productivity and contributing to global food security.

CHAPTER 2

2.1 LITERATURE REVIEW

The paper "Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence" by Natasha Nigar et al. presents a novel show combining profound learning and reasonable AI (XAI) for plant malady location. Given the financial affect of plant illnesses, fast and precise discovery is fundamental, particularly as conventional demonstrative strategies are subjective and time-consuming. Leveraging CNNs, particularly EfficientNetB0, this ponder accomplishes tall classification precision whereas guaranteeing interpretability through the LIME system, which highlights picture locales affecting the model's decisions. The dataset utilized comprises of 87,000 pictures over 14 plant species, classified into 38 malady categories. EfficientNetB0, with 237 layers, accomplished exactness, exactness, and review rates of 99.69%, 98.27%, and 98.26%, separately, beating other state-of-the-art models like MobileNetV2 and ResNet-50. XAI plays a pivotal part here, as LIME gives visual clarifications for forecasts, tending to the "black-box" challenge of profound learning and upgrading believe among end-users. The creators moreover created a versatile app, 'PlantCare,' empowering agriculturists to get to infection expectations and treatment proposals specifically. In spite of the fact that the think about illustrates noteworthy comes about, it recognizes impediments, such as dependence on a single dataset and constrained comparison models. Future investigate may extend illness scope, broaden datasets, and coordinated IoT gadgets for real-time checking. This consider highlights how combining DL with XAI can back straightforward, significant plant malady conclusion, contributing to feasible horticulture and nourishment security.

The paper "Multi-Plant and Multi-Crop Leaf Disease Detection and Classification using Deep Neural Networks, Machine Learning, Image Processing with Precision Agriculture - A Review" examines how to identify plant leaf diseases using machine learning (ML) and deep learning (DL) methods. It discusses how plant diseases have an economic influence on agriculture worldwide, emphasizing how labor-intensive and ineffective old manual diagnosis techniques are for large-scale farms. By precisely detecting infections early on, automated detection with DL and ML can greatly enhance disease management. The paper examines a number of DL models, particularly CNN-

based architectures that have demonstrated good illness classification accuracy, such as EfficientNet, AlexNet, and ResNet. CNN models outperform more conventional ML techniques like SVM and KNN in their ability to capture intricate features in leaf photos. methods for processing images, such as segmentation and K- means clustering and thresholding are popular techniques for feature extraction, which is crucial for identifying sick regions on leaves. To improve model performance, the authors also go over preprocessing techniques including noise reduction and scaling. The quality of the dataset, intra-class variances, and problems with image lighting and scale are among the difficulties identified. The integration of IoT-based biosensors, genetic analysis for pathogen- specific detection, and genetic modification for enhanced crop resilience are among the future directions highlighted in the paper. In order to reduce crop losses and advance agricultural sustainability, the study's insights help develop automated and accurate agricultural solutions.

The paper, "A Mobile-Based Framework for Identifying Plant Leaf Maladies Utilizing Profound Learning," presents a portable application for real-time plant infection discovery utilizing a CNN show. It addresses the require for exact and available malady conclusion, which is basic for lessening rural misfortunes due to plant infections worldwide.traditional illness determination strategies are regularly constrained by assets and precision, particularly for small-scale agriculturists. This think about leverages CNNs to robotize the infection distinguishing proof handle, permitting ranchers to capture pictures of infected takes off and get moment input through a portable app. The demonstrate, prepared on 96,206 pictures over 38 malady categories in 14 trim sorts, accomplished a tall classification precision of 94%. The paper surveys related works, noticing that whereas a few profound learning approaches exist, most need real-time, mobile-ready applications. The framework plan incorporates cloud- based preparing, where CNN classifications are optimized utilizing Intel's Open VINO toolkit, guaranteeing moo latency. In tests, the demonstrate effectively performed malady classification with tall exactness and an normal reaction time of beneath one moment. Future bearings incorporate extending the dataset, making strides show exactness for similar-looking infections, and investigating UAV integration to screen crops over expansive areas. This work gives a down to earth arrangement for field-based illness determination, with noteworthy potential to bolster economical agribusiness.

The paper, "dCrop: A Deep-Learning Based System for Precise Expectation of Maladies of Crops in Savvy Agribusiness," presents a portable app, dCrop, for real-time trim malady location utilizing profound learning. The app addresses the broad issue of trim infections, which cause significant misfortunes in farming, by empowering ranchers to distinguish maladies rapidly and offline utilizing

smartphone cameras. Existing strategies for infection expectation, such as backpropagation systems, Irregular Timberland, K-Means clustering, and Bolster Vector Machines (SVM), have restrictions in terms of overfitting, real-time pertinence, and execution in uncontrolled situations. This ponder makes strides on these with a CNN-based show prepared on the Plant Town dataset, containing 54,306 pictures of 14 crops and 26 maladies. After testing different models, the creators found ResNet 50 to be the most viable, accomplishing 99.24% accuracy. The prepared show was changed over from PyTorch to TensorFlow, permitting for integration into the dCrop app. Agriculturists can utilize this app to capture trim pictures, get moment analyze, and distinguish infections with tall precision, all without requiring web network. The app's offline usefulness and tall execution make it a profitable device for feasible horticulture. Future enhancements recommended incorporate including territorial dialect back and malady administration exhortation.

The paper titled "Deep Crop: Profound Learning-Based Trim Infection Expectation with Web Application" proposes a profound learning system for edit infection discovery and forecast through a web application interface, pointing to help ranchers by distinguishing plant illnesses early for provoke treatment. The consider centres on three profound learning models—CNN, VGG-16, VGG-19, and ResNet-50—evaluating their execution on a 10,000-image dataset comprising takes off from crops like tomatoes, potatoes, and peppers. ResNet-50 outflanked others with an precision of 98.98%, which driven to its usage in the web application. In the broader writing, profound learning has been progressively connected in agrarian settings for plant infection discovery due to its picture classification capabilities. Early works by Sannakki et al. and Jhuria et al. utilized easier machine learning strategies like SVM and ANN, but later headways use profound CNNs to accomplish higher precision. The consider audits approaches that utilize different pre-trained models, such as VGG-16 and Inception-v3, but notes restrictions in exactness and viable applications. Particularly, ResNet-50's engineering, with its more profound organize and capacity to moderate the vanishing angle issue, is famous for its adequacy in making strides classification accuracy. The Deep Crop application, built utilizing Jar and ResNet-50, permits clients to transfer pictures of edit clears out, which the demonstrate classifies as solid or unhealthy. The application not as it were predicts infection nearness but too gives treatment proposals, situating it as a viable device for ranchers. This think about concludes with recommendations for future advancements, counting growing the model's illness scope through bigger datasets and joining localization highlights for more exact infection recognizable proof.

The paper "An Analysis of Deep Learning Techniques for Plant Leaf Disease Detection" by

Sunayana Arya and Rajeev Singh provides an in-depth overview of various deep learning methods applied to plant leaf disease detection. The creators emphasize the significance of opportune illness recognizable proof to relieve trim misfortunes and progress rural efficiency. Conventional strategies for illness discovery are labour-intensive and expensive, provoking the require for more robotized, cost-effective, and exact arrangements. The paper audits the advancement of profound learning approaches, particularly Convolutional Neural Systems (CNNs), which have demonstrated successful for image-based malady classification. A comparative examination highlights CNN models like Alex Net, Google Net, VGGNet, ResNet, and Dense Net. Alex Net's profound structure and utilize of dropout layers oversee overfitting viably, whereas Google Net's beginning modules improve highlight extraction. VGGNet's profundity essentially boosts precision, ResNet addresses the vanishing slope issue with personality mapping, and Dense Net's layer network offers parameter effectiveness for complex images. Limitations incorporate reliance on high-quality, explained datasets and trouble recognizing inside or morphologically comparable infections. Moreover, feebly labelled information decreases exactness, highlighting the require for comprehensive explanations. The creators conclude that extending dataset measure, quality, and differing qualities, along with investigating modern models, seem move forward precision and broaden application in agriculture.

The paper, "Early Detection and Classification of Plant Diseases/Abiotic Disorders using Deep Learning - A Review," Reviews the headways in utilizing profound learning (DL) strategies for early plant malady location, emphasizing its part in boosting rural efficiency and tending to nourishment security concerns. It talks about plant maladies and abiotic clutters, such as those caused by pathogens and natural components, both of which altogether affect trim yields. Traditional machine learning (ML) strategies, whereas profitable, have confinements in manual include extraction, requiring division and tall labour input. In differentiate, DL approaches, particularly convolutional neural systems (CNNs), robotize include extraction, making strides classification exactness for plant maladies. The paper looks at prevalent CNN structures like Alex Net, VGG, ResNet, and Dense Net, highlighting their application in real- world plant illness detection. Additionally, the audit dives into particular strategies such as exchange learning, seriousness investigation, visualization, and include combination. Exchange learning is particularly advantageous in rural applications by adjusting pre-trained models to unused plant datasets, decreasing preparing costs. The survey highlights striking thinks about, such as those utilizing Dense Net for tomato infections and VGG16 for potato illness discovery, accomplishing tall classification correctness's. It

moreover notices commonsense mobile-based applications that utilize lightweight CNN models, permitting agriculturists to distinguish maladies in the field. Challenges incorporate impediments in dataset differences, little injury discovery, and natural inconstancy in real-world scenarios. The paper recommends future investigate bearings, such as improving datasets and joining multispectral imaging for a more exact early location framework, which seem encourage make strides the adequacy of DL in agriculture.

The paper "DeepCrop: Deep Learning-based Crop Disease Prediction with Web Application" presents a system for early location of trim maladies utilizing profound learning models. The think about utilizes CNN, VGG-16, VGG-19, and ResNet-50 models on a dataset of 10,000 plant pictures, centering on crops like tomatoes, peppers, and potatoes. Among these models, ResNet-50 accomplished the most elevated exactness of 98.98%, driving to its integration into a web-based application for real-time trim malady diagnosis. The writing survey highlights past investigate on plant illness discovery utilizing machine learning and profound learning procedures. Thinks about utilizing CNN-based models have detailed correctness's between 93-98%, but these endeavours frequently depended on little datasets and needed real-world appropriateness. Exchange learning models such as Efficient Net and DenseNet too illustrated tall execution but were not optimized for trim malady location or viable deployment. DeepCrop's key commitment lies in its real-world utility. The proposed web application permits agriculturists to transfer photographs of plant clears out for illness determination, advertising treatment proposals. This real-time arrangement addresses the crevice in past investigate by giving a versatile, commonsense instrument for horticulture. The creators recommend future work to incorporate cross breed models and localization procedures for moved forward precision. By and large, DeepCrop improves the utilize of profound learning in farming by advertising a high-accuracy, open arrangement to relieve trim losses.

The paper "Machine Learning for Detection and Forecast of Edit Diseases and Pests: A Comprehensive Overview" audits the application of machine learning (ML) strategies in farming, centering on recognizing, classifying, and foreseeing edit illnesses and bugs. It emphasizes the developing require for robotization due to the impediments of conventional manual infection recognizable proof strategies, especially in huge ranches. The audit examines different ML models, counting Convolutional Neural Systems (CNNs) for picture classification and Long Short-Term Memory (LSTM) systems for anticipating bother events utilizing climate data. The creators highlight the significance of coordination different information sources, such as lackey pictures, climate records, and sensor information, to improve

show exactness. Devices like Plantix and Agrio illustrate how ML can decrease pesticide utilize and make strides cultivate efficiency. In any case, the audit focuses out challenges with information quality, particularly the generalization of models prepared in controlled situations to field conditions. Remote detecting strategies, like Normalized Distinction Vegetation File (NDVI), are appeared to be compelling for observing plant wellbeing and foreseeing episodes. The paper too emphasizes the potential of exchange learning to make strides demonstrate execution when restricted rural datasets are accessible. Future headings incorporate upgrading ML models to handle real-world boisterous information and combining multi-modal datasets for more exact expectations. In general, the survey underscores the transformative part of ML in accomplishing feasible, data-driven cultivating arrangements.

The paper titled "Automation of Crop Disease Detection through Conventional Machine Learning and Deep Transfer Learning Approaches" by Orchi et al. compares machine learning (ML) and profound learning (DL) models for edit illness discovery. Given the rural require for early, exact malady recognizable proof, the ponder utilizes the PlantVillage dataset to classify leaf pictures into solid and infected categories. Key calculations tried incorporate conventional ML models like SVM, KNN, and Gullible Bayes, nearby CNN-based DL models such as InceptionV3 and VGG16. The creators advance improve CNN execution utilizing different actuation capacities (e.g., ReLU, Softmax) and optimizers (SGD, Adam). The discoveries demonstrate that DL, especially InceptionV3, beaten ML models with a best exactness of 98.01%, whereas Credulous Bayes appeared the most reduced precision at 60.09%. This inquire about adjusts with the drift of utilizing profound models for higher exactness in image- based malady location, where DL's capacity to demonstrate complex designs demonstrates beneficial over conventional ML. By differentiating these models and refining CNN arrangements, the think about highlights DL's potential for down to earth, productive, and adaptable trim wellbeing checking. The work gives important bits of knowledge for creating progressed agrarian apparatuses, recommending that DL can considerably move forward illness administration and bolster accuracy agribusiness activities.

The paper "dCrop: A Deep-Learning based Framework for Accurate Prediction of Diseases of Crops in Smart Horticulture" by Pallagani et al. presents a profound learning-based smartphone application for trim infection location. Utilizing ResNet-50, a vigorous CNN engineering, the creators prepared the show on the PlantVillage dataset, containing over 54,000 clarified pictures of sound and infected clears out over 14 trim sorts and 26 illness categories. The show accomplishes a tall precision of 99.24% and is inserted into the dCrop portable app, which empowers offline illness discovery for agriculturists through smartphone cameras. Traditional strategies for edit illness discovery, like Irregular

Timberland, Choice Trees, and K-Means, have impediments such as overfitting, tall parameter affectability, and reliance on web network. dCrop addresses these issues with an optimized, precise demonstrate that runs productively on portable gadgets without web, making it available and commonsense for real- world agrarian settings. Testing on genuine pictures affirmed the app's unwavering quality, with dCrop accurately distinguishing 36 out of 38 diseases. This consider highlights the potential of smartphone-based profound learning applications in supporting maintainable farming. With highlights like offline capability and user-friendliness, dCrop engages ranchers to distinguish infections early, possibly growing to incorporate multilingual bolster and treatment proposals. The apparatus embodies how innovation can upgrade agrarian efficiency and edit assurance.

The paper "Plants Diseases Prediction Framework: An Image-Based System Using Deep Learning" by Madhu Kirola and colleagues presents a system for recognizing plant maladies utilizing machine learning (ML) and profound learning (DL) methods. The creators note that plant illnesses basically influence clears out, affecting edit quality and abdicate. Their proposed system incorporates a few image-processing steps: securing, pre-processing, division, include extraction, and classification. The infections are classified as bacterial, contagious, or viral, utilizing both conventional ML classifiers and a profound learning demonstrate (CNN). In the writing audit, the creators summarize existing works on plant illness discovery. Md. Asif et al. utilized picture division with ML classifiers, finding Irregular Timberland (RF) to be the most exact. Vyshnavi utilized SVM for contagious illness classification, whereas Arivazhagan utilized HSI color models and surface highlights with SVM for illness discovery. Other considers investigated profound learning approaches: Nage utilized CNNs for identifying common plant maladies, whereas Francis et al. accomplished 88.7% precision in a parallel classification of tomato and apple leaf wellbeing utilizing CNNs. The authors' exploratory comes about uncover that, among the ML models tried, RF accomplished the most elevated exactness (97.12%). Be that as it may, CNN beat RF with an precision of 98.43%. The think about highlights CNN's quality in dealing with huge datasets with tall precision, whereas conventional ML models may be more resource-efficient but less exact. The paper concludes that future inquire about seem include applying picture enlargement and real-time execution of these models for broader agrarian applications.

The paper "Plant Disease Detection Using Deep Learning" by Kowshik B. et al. explores the use of deep learning models, especially Convolutional Neural Systems (CNNs), for identifying and classifying plant illnesses from leaf pictures. The creators emphasize the centrality of horticulture to the Indian economy and the challenges postured by plant infections, which diminish both trim abdicate and quality.

Conventional strategies depend intensely on human skill and visual assessment, which are time-consuming, expensive, and illogical for large-scale cultivating. Subsequently, the creators propose a profound learning-based computerized framework for early detection. In their writing audit, the creators examine past works that apply machine learning and picture preparing to plant illness discovery. For occurrence, Gaurav Verma et al. utilized CNN for rice malady location, whereas Nikhil Shah and Sarika Jain connected fake neural systems to classify maladies in cotton takes off. Another ponder by Sardogan et al. coordinates CNN with Learning Vector Quantization for leaf malady location in plants. These earlier approaches illustrate the guarantee of neural systems in identifying trim maladies but highlight the require for more prominent precision and cost-effectiveness. The proposed strategy in this paper employs picture procurement, preprocessing, improvement, division, and include extraction. Key highlights such as color and surface are extricated, and CNNs analyze these highlights to classify illnesses in plant takes off. This framework permits effective preparing of picture information through OpenCV, which empowers real-time malady classification utilizing a low-cost camera setup. The creators conclude that CNN and Profound Neural Organize (DNN) structures are compelling for plant infection discovery and can enormously help agriculturists by giving early determination, in this way minimizing trim misfortune. The paper recommends that future work might extend the system's usefulness to offer extra administrations, such as nearby advertise costs and government assets, assist helping ranchers.

The paper titled "Crop Disease Prediction using Machine Learning and Deep Learning: An Exploratory Study" addresses the basic issue of trim malady location, basic for keeping up edit surrender and quality, especially on large-scale ranches. Conventional strategies like visual assessment are labor-intensive and inclined to human blunder, underscoring the require for robotized, versatile arrangements. The consider audits later progressions in Machine Learning (ML) and deep Learning (DL) for edit illness discovery, highlighting the victory of Convolutional Neural Systems (CNNs) in distinguishing maladies from leaf pictures. CNNs are especially famous for their include extraction capabilities, which permit precise infection classification indeed with complex visual patterns. The paper moreover investigates cross breed models that combine ML and DL, such as SVM-CNN systems, which capitalize on both methods' qualities to move forward precision and flexibility. Moreover, exchange learning has appeared guarantee in adjusting pre-trained models to modern trim datasets, making it an productive choice for resource-limited setups. The consider underscores the significance of high-quality datasets, like the Plant Town dataset, but notes a hole in different real-time information from different natural conditions. Challenges stay in improving demonstrate strength and extending datasets to cover

more crops and conditions. Future headings include coordination ML/DL with IoT and portable innovations for real-time observing, which seem give convenient malady administration suggestions for agriculturists. In general, this think about highlights the potential of progressed AI strategies to change trim malady administration, advancing effective, exact, and adaptable arrangements in agriculture.

The paper “Machine Learning Techniques for Plant Infection Detection: An Evaluation with a Customized Dataset” addresses challenges in plant illness discovery utilizing profound learning (DL) strategies. Plant illnesses affect nourishment security and agriculturist employments, however current location strategies are regularly costly or blocked off. This consider presents a customized dataset of 87,570 leaf pictures, crossing 32 plant species and 74 infections, to bolster more vigorous DL models. The creators assess four convolutional neural arrange (CNN) models—VGG-16, VGG-19, ResNet-50, and ResNet-101—using exchange learning, which leverages pretrained ImageNet weights for improved performance. The think about finds that VGG-16 and VGG-19 beat ResNet models, accomplishing precision around 86% and F1-scores of 87%, recommending these models are way better suited for plant illness discovery. In comparison, ResNet-50 and ResNet-101 battled, particularly with real- world conditions. The dataset made interestingly incorporates pictures from both lab- controlled and real-world settings, tending to a major restriction in earlier datasets that depend on controlled environments. The creators highlight that utilizing VGG models in genuine rural settings can make strides location exactness and help in early mediation, decreasing financial misfortunes. Future work includes adjusting dataset classes and investigating more progressed methods like generative antagonistic systems (GANs) to advance improve demonstrate execution. This paper makes critical commitments by presenting a dataset optimized for reasonable applications and approving CNN structures reasonable for viable, computerized plant infection detection.

The paper "Enhancing Crop Recommendation Frameworks with Explainable Artificial Intelligence" presents XAI-CROP, a edit suggestion demonstrate utilizing logical AI (XAI) to upgrade straightforwardness in decision-making for agriculturists. Conventional edit suggestion frameworks (CRS) frequently work as "dark boxes," restricting believe due to their need of interpretability. This investigate builds on earlier considers appearing that ML procedures can anticipate trim yields by leveraging soil, climate, and chronicled information, but it extraordinarily joins XAI to make proposals more understandable. The creators survey past work where models like Convolutional Neural Systems (CNNs) and Repetitive Neural Systems (RNNs) progress edit forecast by utilizing farther detecting and natural information. Other approaches, such as the Trim Abdicate Expectation Calculation (CYPA),

utilize IoT information to make strides in surrender figures. The XAI-CROP demonstrate utilizes choice trees nearby the Nearby Interpretable Model-Agnostic Clarifications (LIME) method to clarify its predictions. XAI-CROP was tried against models like Angle Boosting and Irregular Timberland, accomplishing higher interpretability and exactness. It had lower Cruel Squared Blunder (MSE) and Cruel Supreme Mistake (MAE), as well as a higher R-squared (R²), appearing solid prescient capacity. By improving CRS straightforwardness, XAI-CROP may increment selection in farming, possibly supporting maintainable cultivating hones and nourishment security. This inquire about contributes to the developing body of writing supporting XAI in commonsense, data-driven agribusiness.

Paper	Model/Technique	Dataset	Accuracy/Performance	Limitations	Future Directions
Improving Plant Disease Classification with DL + XAI	CNN (EfficientNetB0) with LIME	87,000 images, 14 plant species, 38 diseases	Accuracy: 99.69%, Precision: 98.27%, Recall: 98.26%	Limited to single dataset, few comparison models	Expand disease scope, IoT integration for real-time monitoring
Multi-Plant and Multi-Crop Leaf Disease Detection and Classification	CNN-based architectures (EfficientNet, etc.)	ML and DL models, uses image processing for feature extraction	High classification accuracy	Dataset quality, intra-class variance, image lighting issues	IoT-based biosensors, genetic analysis, enhanced crop resilience
Mobile-Based Framework for Identifying Plant Leaf Diseases	CNN, optimized with Intel's OpenVINO	96,206 images, 38 disease categories, 14 crop types	Accuracy: 94%	Limited accuracy with similar-looking diseases, no UAV integration	Expand dataset, improve model accuracy, explore UAV integration
dCrop: Real-time Crop Disease Detection App	ResNet-50	PlantVillage, 54,306 images, 14 crops, 26 diseases	Accuracy: 99.24%	Offline function limited to pre-trained classes	Add regional language support, disease management advice

DeepCrop: DL-based Crop Disease Prediction with Web Application	CNN, VGG-16, VGG-19, ResNet-50	10,000 images (e.g., tomatoes, potatoes, peppers)	Highest accuracy: 98.98% with ResNet-50	Needs larger datasets, lacks localization features	Expand model's disease coverage, add localization for precise diagnosis
An Analysis of DL Techniques for Plant Leaf Disease Detection	CNNs (AlexNet, GoogleNet, VGGNet, ResNet)	Varied datasets, focuses on detailed comparisons between CNN models	Comparisons of multiple architectures	High dependency on quality-labeled datasets, hard to identify similar diseases	Improve dataset variety, explore new architectures for broader applications
Early Detection of Plant Diseases Using Deep Learning	CNN models (AlexNet, VGG, ResNet)	Large datasets focusing on various crops	High classification accuracy	Dataset diversity, limited in-field variability	Enhance datasets, integrate multispectral imaging for early detection
Machine Learning for Crop Disease and Pest Forecasting	CNN for images, LSTM for pest prediction	Integrates image data, weather, and sensor inputs	Reduction in pesticide use, improved farm productivity	Data quality issues, model generalization in field conditions	Refine ML for noisy real-world data, combine multimodal datasets for accurate predictions
Automation of Crop Disease Detection with ML and DL	ML (SVM, KNN, Naïve Bayes), CNN	PlantVillage dataset for classification	Highest accuracy: 98.01% with InceptionV3	Lower accuracy with Naïve Bayes, less suitable for complex image data	Further refine CNN performance, test in real-world scenarios for practical

					crop monitoring
Plants Disease Prediction with Deep Learning (DeepCrop)	CNN, with a web app for real-time diagnosis	10,000 plant images (e.g., tomatoes, peppers)	Accuracy: 98.98% with ResNet-50	Limited real-world applicability due to small datasets	Explore hybrid models, add localization for better accuracy
Machine Learning for Crop Disease Detection: Customized Dataset Evaluation	CNN (VGG-16, VGG-19, ResNet-50, ResNet-101)	87,570 images, 32 plant species, 74 diseases	Best accuracy ~86%, F1-score ~87% with VGG models	Difficulty generalizing to new plant diseases and crop types	Investigate GANs for class balance, expand to other plant species
Enhancing Crop Recommendation with Explainable AI (XAI-CROP)	Decision Trees with LIME	Crop data from India, focusing on soil, weather, and historical crop yields	Lowest MSE: 0.9412, MAE <1, R2: 0.94152	Region-specific dataset, scalability and data access issues	Improve dataset diversity, optimize model for other regions, refine explanations

Table 2.1

2.2 LIMITATIONS OF THE EXISTING SYSTEM

- **Single Dataset Dependency:** Many studies rely on a single dataset, like "Plant Village," which may limit the model's ability to generalize in diverse real-world conditions with variable environmental factors.
- **Limited Model Comparisons:** Some studies only compare their model to a few pre-trained models (e.g., MobileNetV2, ResNet-50), missing a broader range of advanced architectures that could offer more comprehensive performance assessments.
- **Dataset Imbalance:** Certain studies use imbalanced datasets, with underrepresented classes, which could bias the model's performance and reduce accuracy for less-represented diseases.
- **Fixed Model Architecture:** By focusing on a specific architecture (e.g., EfficientNetB0), models may lack flexibility, making it harder to adapt to various computational resources or dataset sizes.
- **Lack of Real-World Testing:** Many studies report high accuracy in controlled experimental settings but lack extensive real-world validation, which is essential to assess model robustness in varied agricultural environments.
- **Reliance on LIME for Interpretability:** While LIME provides local interpretability, it doesn't offer a global view of model behavior. Additional XAI techniques, like SHAP or Grad-CAM, could improve overall interpretability.
- **Absence of IoT Integration:** Although IoT integration for real-time monitoring is proposed, it's not implemented in many studies, limiting the system's utility for continuous disease tracking in the field.
- **Resource Requirements:** Models like EfficientNetB0 and XAI techniques such as LIME demand high computational resources, making them less accessible for small-scale farmers with limited technology.
- **Controlled Dataset Bias:** Models trained on datasets like PlantVillage may not perform well in real-world conditions where variables like lighting, background, and leaf orientation vary widely.
- **Dataset Constraints:** Many datasets used don't encompass the full diversity of plant species and disease types, restricting the models' applicability across different crops and regions.
- **Computational Resource Constraints:** CNN-based deep learning models require significant computational resources, which might be a limitation for small-scale farmers lacking access to high-end hardware.
- **Emphasis on Image-Based Detection:** Most studies focus on image-based disease detection,

ignoring other methods (e.g., genetic analysis, biosensors) that could complement visual approaches for better accuracy.

- **Lack of Field-Based Validation:** The models are primarily validated in lab settings rather than actual agricultural fields, where conditions are less predictable.
- **Difficulty with Multiple Disease Detection:** Most models struggle with identifying multiple diseases on a single leaf, reducing their effectiveness in complex real-world scenarios.
- **Economic Feasibility Unaddressed:** Many studies lack analysis on the economic feasibility of deploying these technologies, which is essential for resource-limited farmers.
- **Environmental Impact Overlooked:** Few studies consider the environmental implications of large-scale technology deployment, such as increased electronic waste from IoT sensors.
- **Big Data Management Challenges:** Handling and managing the large datasets required for these models is not addressed, which is crucial for scaling these systems.
- **Scalability Issues:** Limited insight into how these models could scale to smallholder versus large-scale farms, which have varying technological and infrastructural needs.
- **Internet Connectivity Dependency:** Cloud-based processing often requires a stable internet connection, which may be unreliable in rural areas, restricting model usability.
- **Device Compatibility and Accessibility:** Some applications are only available on specific platforms (e.g., Android), limiting access for users on other devices and platforms, such as iOS.
- **Poor Adaptability to Extreme Environmental Conditions:** Many models are trained under controlled conditions, making them less adaptable to extreme weather or lighting variations commonly found in real-world agricultural settings.
- **Generalization to Unseen Crop Types:** Models trained on specific crop types may not generalize well to new crops, limiting their usefulness in regions with different agricultural focuses.
- **Single-Disease Detection Limitation:** Most models are only capable of identifying one disease per image, which does not reflect real-life situations where multiple diseases may affect a plant simultaneously.
- **Dependence on High-Quality Images:** The accuracy of disease detection relies heavily on clear, high-quality images, making it difficult for models to perform well with poor image quality or occlusions (e.g., shadows, partial leaves).
- **No Support for Disease Localization:** Some models lack localization features to pinpoint affected areas on the leaf, which could provide more precise recommendations for treatment.

- **Potential Overfitting to Training Data:** Models with high accuracy on training datasets may be overfitted, limiting their performance on new or varied images.
- **Lack of Multilingual Support:** Applications are often only available in a single language (e.g., English), which restricts access for non-English-speaking farmers, especially in rural areas.
- **Absence of Recommendations for Disease Management:** Many systems identify diseases but do not provide actionable guidance on disease treatment or pesticide use, which would be highly valuable to farmers.
- **Storage and Maintenance Requirements for Big Data:** The need for extensive data storage and management resources for large datasets isn't often addressed, which is crucial for scaling these solutions in data-intensive applications.
- **Environmental Variability Not Considered:** Key environmental factors like soil type, seasonality, and humidity levels are often overlooked, yet they play a significant role in disease occurrence and could affect detection accuracy.

2.3 SCOPE OF THE PROJECT

1. Objective and Application

The essential scope of this extend is to create a solid profound learning-based framework for crop illness classification utilizing a convolutional neural arrange (CNN). This venture can be connected in rural settings to help ranchers, analysts, and agronomists in recognizing plant illnesses expeditiously and precisely. Early location can altogether move forward edit administration and diminish misfortunes due to disease.

2. Information Utilization and Preprocessing

- **Dataset:** The venture employments the PlantVillage dataset, which contains a different set of pictures of sound and ailing plant takes off. This dataset empowers preparing a vigorous demonstrate that can generalize well to modern, inconspicuous data.
- **Preprocessing:** Information preprocessing incorporates picture normalization and increase (e.g., rescaling and even flipping). These steps improve the preparing prepare by avoiding overfitting and guaranteeing that the demonstrate can learn invariant highlights from diverse picture perspectives.

3. Demonstrate Development

- **CNN Engineering:** The extend utilizes a custom-built CNN with numerous convolutional and pooling layers planned to capture complex designs and highlights in the input pictures. This custom-made show design points for tall exactness over 38 plant illness categories.
- **Training Arrangement:** The demonstrate preparing includes hyperparameter tuning with a set number of ages, a group estimate of 32, and execution optimization utilizing Adam as the optimizer. Callbacks like ModelCheckpoint and EarlyStopping are actualized to screen preparing and anticipate overfitting.

4. Show Evaluation

- **Performance Measurements:** The extend assesses the demonstrate utilizing standard measurements such as exactness, AUC, and a disarray network. The consideration of these measurements makes a difference in understanding the model's qualities and shortcomings over distinctive classes.
- **Classification Report:** Nitty gritty assessment with accuracy, review, and F1-score gives experiences into class-wise execution, permitting for alterations in preparing or show design if needed.

5. Explainability with LIME

- **Model Interpretability:** A noteworthy portion of this project's scope includes joining LIME (Neighborhood Interpretable Model-agnostic Clarifications) for clarifying the model's forecasts. LIME makes a difference to recognize which parts of an input picture most impact the model's choice. This perspective includes straightforwardness to the profound learning demonstrate, making it simpler for non-technical partners to believe and get it how choices are made.
- **User Engagement:** The explainability component is especially important for clients who require to confirm the model's behavior, such as distinguishing whether it centers on significant picture highlights like spots, discolorations, or leaf shape.

6. Potential Extensions

- **Model Improvements:** Future emphasess of the extend may incorporate fine-tuning with pre-trained models such as VGG16 or ResNet for possibly progressed performance.
- **Real-Time Applications:** Amplifying the venture to incorporate a portable app or a web interface where clients can transfer leaf pictures and get real-time forecasts and explanations.
- **Scalability:** Extending the dataset to incorporate more plant species and maladies to make a more comprehensive show appropriate to assorted agrarian needs.

7. Challenges and Considerations

- **Data Inconstancy:** Guaranteeing that the show generalizes well requires overseeing information quality and potential predispositions in the preparing set.
- **Explainability Trade-offs:** Whereas LIME gives interpretability, it may include computational overhead and may not be as exact as model-specific explainability methods.
- **Deployment:** Contemplations for conveying the demonstrate in resource-constrained situations (e.g., portable gadgets) might require optimization procedures like show pruning or quantization.

CHAPTER 3

SYSTEM DESIGN

3.1 PROBLEM DEFINITION

Agriculture is considered a crucial sector globally, as it plays a significant role in food production and supports rural economies. Crop diseases pose a significant challenge, causing substantial yield losses and endangering food security. Traditional methods of identifying crop diseases heavily depend on the knowledge of agronomists and plant pathologists. This makes timely intervention challenging, especially in remote and resource-limited areas. Two Moreover, the increasing intricacy of disease variations and environmental influences necessitates scalable, data-centric methods to precisely anticipate and identify crop diseases.

Objective

The project aims to create a machine learning model that can predict crop diseases using input data like images, weather conditions, soil information, and crop health indicators. Achieving high accuracy is certainly crucial as it is vital for farmers and agricultural experts to comprehend the model's predictions. Hence, explainability plays a crucial role by allowing stakeholders to confidently understand, interpret, and implement the predictions generated by the model.

Scope of the Problem

- **Data Availability and Types:**

High-quality labeled data on crop diseases, encompassing visual data such as leaf, stem, and plant images, as well as environmental data like weather and soil conditions. Multimodal data integration presents challenges, given the wide variation in format, quality, and sources of agricultural data.

- **Accuracy vs. Interpretability:**

Standard deep learning models, such as Convolutional Neural Networks (CNNs), deliver high accuracy in image classification yet are often perceived as "black boxes," lacking interpretability. The project demands a delicate equilibrium between achieving superior predictive accuracy and ensuring that the outcomes are comprehensible for end-users.

- **User Understanding and Trust:**

Farmers and agricultural practitioners could have limited technical knowledge. Hence, forecasts need to be communicated in a way that is both clear and easy to act upon. Providing clear

explanations that pinpoint the specific areas of the input data, such as regions within an image, that influenced the prediction, can foster trust and help in making informed decisions.

Objectives

- **Disease Prediction Model:** Create a resilient model capable of accurately forecasting the existence and types of crop diseases. **Explainability:** Enhance transparency in AI systems by using explainable AI (XAI) methods such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) to demonstrate the significance of features in tabular data. Explore Grad-CAM, a method that offers visual explanations for image-based predictions. It highlights the critical regions of an image that influenced the decision-making process. Create a user-friendly interface by designing a dashboard or mobile application that presents predictions and explanations in a simplified manner. This interface should be easily accessible to farmers and agricultural advisors.

3.2 SYSTEM ARCHITECTURE

1. Outline of Code Structure and Purpose

The objective of the given code is to create a convolutional neural network (CNN) for classifying plant infections from picture information. The code is organized to include:

- **Data Dealing with:** Stacking and organizing picture information into organized formats.
- **Preprocessing:** Planning pictures through increase and normalization, and part information into preparing, approval, and test sets.
- **Model Development:** Characterizing the CNN design with convolutional and pooling layers.
Training: Compiling and preparing the show, utilizing callbacks for optimization.
- **Evaluation:** Surveying the model's execution utilizing different metrics.
Explainability: Utilizing LIME (Neighborhood Interpretable Model-agnostic Clarifications) to make the model's expectations more interpretable.

2. Information Taking care of and Preprocessing

- **Data Stacking:** The information() work peruses picture record ways and names into a DataFrame.
- **Dataset Get to:** Kaggle API is utilized to download the dataset, which is at that point extricated and organized into subfolders by lesson labels.
- **Data Part:** The dataset is isolated into preparing (80%), approval (10%), and test (10%) sets utilizing train_test_split, with stratification to protect course distributions.
- **Augmentation:** The ImageDataGenerator lesson is arranged to apply picture expansion (e.g., rescaling and even flipping) to increment inconstancy in the preparing set, advancing way better generalization.

3. Demonstrate Engineering and Layers

Layers Configuration

- **Convolutional Layers:** Six layers with 32 and 64 channels, each with a 3x3 part and ReLU activation.
- **Pooling Layers:** Six max-pooling layers to decrease dimensionality and extricate key features.
- **Flatten Layer:** Changes 3D highlight maps into 1D vectors.
- **Dense Layers:** Incorporates one thick layer with 64 units and an yield layer with 38 units (one per lesson) utilizing SoftMax activation.
- **Design Basis:** The design is planned to capture complex picture highlights through numerous layers, empowering vigorous classification.

4. Preparing Setup and Hyperparameters

- **Optimizer and Misfortune Work:** Employments Adam optimizer and categorical_crossentropy for multi-class classification.
- **Batch Estimate:** 32
- **Epochs:** At first set to 5, with the potential for early stopping.
- **Metrics:** Screens precision and AUC (Region Beneath the Curve).
- **Model Checkpoint:** Spares the best demonstrate amid training.
- **Early Stopping:** Stops preparing when approval execution ceases to make strides after 5 ages, diminishing overfitting.

5. Explainability with LIME

- **Purpose:** LIME is utilized to give neighborhood clarifications for person expectations, making the model's decision-making prepare more transparent.
- **Integration:**
Method: LIME creates clarifications by irritating input pictures and analysing how change influence predictions.
- **Steps:**
Select a test picture from the test set.
Use lime image module to make clarifications for the CNN's output.
Highlight locales in the picture that contribute most to the anticipated class.
- **Outcome:** LIME's visual clarifications offer assistance distinguish which picture highlights (e.g., leaf spots or discoloration) are compelling in the model's classification, improving interpretability.

6. Assessment Measurements and Comes about Analysis

- **Model Assessment:** The prepared show is surveyed utilizing measurements such as test misfortune, precision, and AUC.
- **Confusion Network:** A heatmap of the disarray framework is produced to visualize the model's classification execution over all classes.
- **Classification Report:** A point by point report incorporates accuracy, review, and F1-score for each class.
- **Sample Forecasts:** Irregular test expectations are visualized to compare genuine vs. anticipated labels.
- **LIME Examination:** Clarifications from LIME give bits of knowledge into particular locales of

input pictures that the demonstrate depends on, displaying how expectations are made for superior straightforwardness.

Simplified Image Classification Model Training

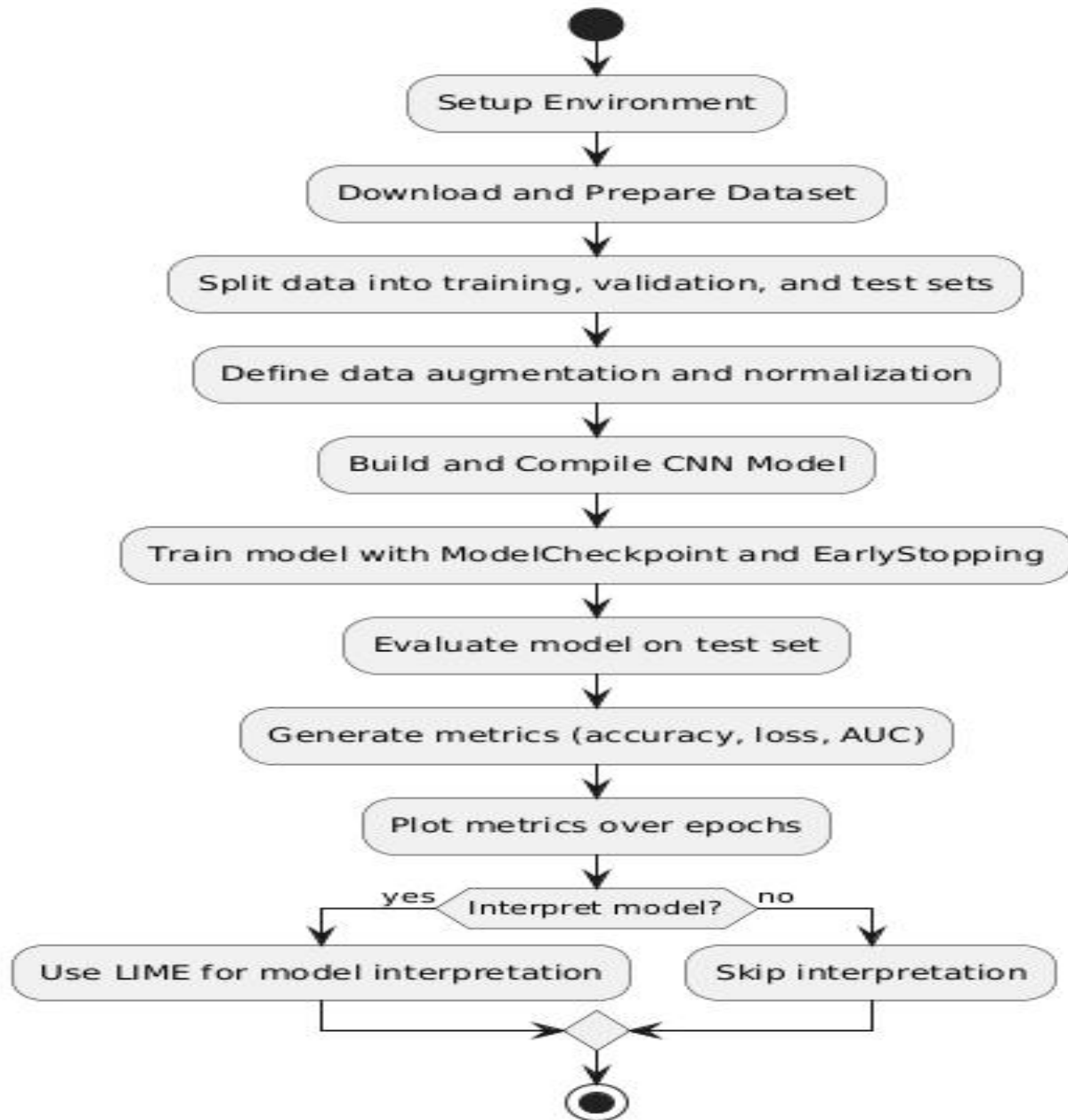


Fig 3.2

3.3 REQUIREMENT SPECIFICATIONS

3.3.1 Software Requirements

1. Programming Language

- **Python:**

The primary programming language for this project due to its rich ecosystem of libraries for data science, machine learning, and deep learning. Python offers simplicity and versatility, making it ideal for rapid prototyping and implementation of deep learning models.

2. Integrated Development Environment (IDE) / Code Editor

- **Jupyter Notebook:** Recommended for iterative coding and interactive data exploration. It supports visualization directly within the notebook, which is useful for displaying images, charts, and model outputs.
- **VS Code or PyCharm:** These IDEs offer advanced debugging, version control integration, and plugin support for seamless code development and management.
- **Google Colab:** A cloud-based alternative that provides free GPU access, which is beneficial for training deep learning models without needing a high-spec local machine.

3. Deep Learning Frameworks

- **TensorFlow:** An open-source platform developed by Google, highly suitable for building and training deep learning models. It supports operations on both CPUs and GPUs, facilitating faster training times.
- **Keras (Integrated with TensorFlow):** Provides a high-level API for building deep learning models. Keras simplifies the process of creating complex neural network architectures with minimal code.

4. Machine Learning Libraries

- **scikit-learn:** Used for tasks like splitting the dataset into training and testing sets (`train_test_split`), calculating evaluation metrics, and creating the confusion matrix and classification report.
- **LIME (Local Interpretable Model-agnostic Explanations):** Used for explainability. LIME helps in understanding the CNN model's decisions by providing visual explanations for predictions.

5. Image Processing Libraries

- **OpenCV (cv2):** Utilized for image loading, processing, and manipulation. This library provides efficient methods to read images in different formats and convert them for use in training (e.g.,

converting color spaces).

- **Pillow (PIL):** An alternative image processing library that can be used alongside or in place of OpenCV for simple image manipulations.

6. Data Handling and Analysis Libraries

- **Pandas:** Used to create and manage dataframes that hold file paths, labels, and other structured data. This simplifies operations like data exploration and preprocessing.
- **NumPy:** A fundamental package for numerical computing, providing support for arrays and matrices, which are essential for data transformations and calculations.

7. Visualization Libraries

- **Matplotlib:** Essential for creating plots to visualize training/validation loss and accuracy over epochs, as well as for displaying sample images and prediction results.
- **Seaborn:** Used for creating more aesthetically pleasing and informative plots, such as confusion matrices with heatmaps.

8. Data Augmentation and Preprocessing

- **ImageDataGenerator (Keras):**__Helps in real-time data augmentation by applying transformations such as rotation, flipping, rescaling, and zooming. This enhances the model's ability to generalize by training on a varied dataset.

9. Model Checkpointing and Callbacks

- **Keras Callbacks (ModelCheckpoint, EarlyStopping):**_Used for saving the best model during training and halting training when no improvement is detected, preventing overfitting and saving computational resources.

10. Environment and Platform Tools

- **Kaggle API:** Allows for seamless data downloading from Kaggle datasets. Requires Kaggle. Json for API authentication.
- **Jupyter Widgets (tqdm):** Provides progress bars for loops to enhance visualization of long-running operations.
- **Operating System Compatibility:** The code should be compatible with Windows, macOS, or Linux-based systems. Running the code on a Linux-based OS (e.g., Ubuntu) is often more efficient for deep learning projects due to better support for GPU drivers and libraries.

11. GPU Support for Training

- **CUDA Toolkit:** Necessary for GPU acceleration with NVIDIA GPUs. CUDA allows TensorFlow

and Keras to offload computationally intensive operations to the GPU.

- **cuDNN:** A deep neural network library by NVIDIA that works with CUDA to optimize performance for deep learning operations.

CHAPTER-4

SYSTEM IMPLEMENTATION

4.1 OVERVIEW OF THE MODULES

The system for plant disease classification is composed of several interconnected modules, each serving a specific function within the machine learning pipeline. The primary modules in this system are:

Data Loading and Preprocessing Module:

- This module involves importing libraries, setting up paths, and creating data structures to read and organize image data. The main goal is to prepare a DataFrame containing image paths and corresponding labels.
- Responsible for loading image data from a structured directory, creating image paths, and preparing labels for classification.

Data Augmentation and Generator Module:

- To train the CNN effectively, this module handles image augmentation and generates batches for training, validation, and testing. Data augmentation improves the model's ability to generalize by introducing variability in the training set.
- Enhances the training data by applying transformations and creating image batches suitable for the training process.

Model Architecture Module:

- The core of the project lies in this module, where a custom CNN model is defined. The architecture is designed to extract spatial hierarchies in image data for accurate classification.
- Defines the Convolutional Neural Network (CNN) architecture tailored for plant disease classification.

Model Training and Validation Module:

- This module focuses on training the model using the prepared data and configuration settings. It includes setting up callbacks to save the best model and prevent overfitting.
- Handles the training of the CNN model, including setting up training callbacks and monitoring performance.

Evaluation and Visualization Module:

- After training, the model is evaluated on the test set, and results are visualized to understand the model's performance better.

- Evaluates the trained model using test data and visualizes the results through confusion matrices and other performance metrics.

Explainability Module:

- The explainability module utilizes LIME to provide interpretable explanations for individual predictions, making the model's behavior more transparent.
- Uses LIME (Local Interpretable Model-agnostic Explanations) to generate insights on how the model makes predictions.

4.2 DESCRIPTION OF THE MODULES

4.2.1 MODULE 1 DESCRIPTION

Data Loading and Preprocessing

Detailed Description:

- **File Path Extraction:** The `data()` function reads image files from the given dataset directory and extracts the paths and labels. It iterates through subdirectories, each representing a class (i.e., plant type and disease).
- **Data Frame Creation:** The collected paths and labels are stored in a Pandas DataFrame, facilitating easy data manipulation and exploration.
- **Key Libraries Used:**
 - `os` for directory traversal.
 - `cv2` for image reading and preliminary processing.
 - `pandas` for creating and managing structured data.

- **Code Breakdown:**

```
def data(dataset_path):  
    images = []  
    labels = []  
    for subfolder in tqdm(os.listdir(dataset_path)):  
        subfolder_path = os.path.join(dataset_path, subfolder)  
        for image_filename in os.listdir(subfolder_path):  
            image_path = os.path.join(subfolder_path, image_filename)  
            images.append(image_path)  
            labels.append(subfolder)  
    df = pd.DataFrame({'image': images, 'label': labels})  
    return df
```

Purpose: This module ensures that image data is well-organized for input into the deep learning pipeline, maintaining class labels for training and evaluation.

Data Augmentation and Generator Module

Detailed Description:

- **ImageDataGenerator:** The ImageDataGenerator class from Keras is utilized for rescaling pixel values and augmenting images through transformations like flipping and rotation.
- **Batch Generation:** The generator creates batches from the DataFrame created earlier, allowing the model to process data efficiently during training and validation.

Augmentation Techniques:

- **Horizontal Flipping:** Randomly flips images to create variations.
- **Rescaling:** Normalizes pixel values to a range of [0, 1].
- **Data Splitting:** The dataset is split into training, validation, and test sets using `train_test_split()` to ensure the model has distinct data for different stages of training and evaluation.

Code :

```
python
train_datagen = ImageDataGenerator(
    rescale=1./255,
    horizontal_flip=True
)
train_generator = train_datagen.flow_from_dataframe(
    df_train,
    x_col='image',
    y_col='label',
    target_size=image_size,
    batch_size=batch_size,
    shuffle=True
)
val_generator = datagen.flow_from_dataframe(
    df_val,
    x_col='image',
    y_col='label',
    target_size=image_size,
```

```
batch_size=batch_size,  
shuffle=False  
)
```

Purpose: This module ensures that the model sees varied versions of the training data, helping it learn more robust features and preventing overfitting.

Model Architecture Module

Detailed Description:

Layer Composition:

- **Convolutional Layers:** Extract features by applying convolution operations to the input images.
- **MaxPooling Layers:** Reduce dimensionality while retaining important information.
- **Dense Layers:** Act as fully connected layers that combine extracted features for classification.
- **Activation Functions:** The relu (Rectified Linear Unit) activation function is used for hidden layers to introduce non-linearity, while softmax is used in the output layer for multi-class classification.
- **Input Shape:** The input layer is defined to accept images of size (224, 224, 3).

Code :

```
model = keras.models.Sequential([  
    keras.layers.Conv2D(32, kernel_size=(3, 3), activation='relu', input_shape=(224, 224, 3)),  
    keras.layers.MaxPooling2D(2, 2),  
    keras.layers.Conv2D(64, kernel_size=(3, 3), activation='relu'),  
    keras.layers.MaxPooling2D(2, 2),  
    # Additional convolutional and pooling layers  
    keras.layers.Flatten(),  
    keras.layers.Dense(64, activation='relu'),  
    keras.layers.Dense(38, activation='softmax')  
])
```

Purpose: This module constructs a deep learning model that can learn complex features and relationships within the input data to classify plant diseases effectively.

4.2.2 MODULE 2 DESCRIPTION

Model Training and Validation Module

Detailed Description:

Loss Function and Optimizer: The model is compiled with categorical_crossentropy as the loss function and Adam optimizer for efficient gradient-based optimization.

Callbacks:

ModelCheckpoint: Saves the best-performing model during training.

EarlyStopping: Stops training early if the validation loss does not improve after a set number of epochs.

Training Execution: The fit() method is called to start training, using the train_generator for input data and val_generator for validation.

Code :

```
checkpoint = ModelCheckpoint("my_keras_model.keras", save_best_only=True)
early_stopping_cb = EarlyStopping(patience=5, restore_best_weights=True)
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy',
keras.metrics.AUC()])
hist = model.fit(train_generator, epochs=5, validation_data=val_generator, callbacks=[checkpoint_cb,
early_stopping_cb])
```

Purpose: This module ensures that the model trains effectively while saving the best version and preventing overfitting through early stopping.

Evaluation and Visualization Module

Detailed Description:

Metrics:

Accuracy, Loss, and AUC: Evaluated using model.evaluate().

Confusion Matrix: Visualized using Seaborn for class-wise performance analysis.

Classification Report: Provides a comprehensive summary of precision, recall, and F1-score for each class.

Graphical Representations:

Training History: Line plots showing training and validation loss, accuracy, and AUC over epochs.

Sample Predictions: Displays a set of images along with their true and predicted labels for visual validation.

Code:

```
score, acc, auc = model.evaluate(test_generator)
print('Test Loss =', score)
print('Test Accuracy =', acc)
print('Test AUC =', auc)
# Plot training and validation metrics
plt.figure(figsize=(25, 10))
plt.subplot(1, 3, 1)
plt.plot(hist_['loss'], 'b-o', label='Train_Loss')
plt.plot(hist_['val_loss'], 'r-o', label='Validation_Loss')
plt.legend()
```

Purpose: This module provides insights into model performance and helps in diagnosing areas for potential improvement.

Explainability Module**Detailed Description:****LIME Integration:**

LIME Image Explainer: Generates explanations by perturbing the input image and observing changes in model predictions.

Explanation Visualization: Highlights parts of the image that contribute the most to a specific prediction, allowing users to validate the model's focus.

Use Case: This is particularly valuable for understanding whether the model correctly identifies disease-relevant regions in an image, enhancing user trust in the model.

CHAPTER-5

RESULTS AND DISCUSSION

In this section, we offer a thorough overview of the model's performance, evaluation, and its implications. The aim of this section is to convey the findings from the CNN model for plant disease classification, thoroughly examine the results, and address any potential enhancements or hurdles encountered in the process.

5.1 DESCRIPTION

Model Performance:

After training the Convolutional Neural Network (CNN) model on the plant disease dataset, several key performance metrics were evaluated to assess the effectiveness of the model in classifying plant diseases based on image data. The main criteria considered for evaluation were:

- **Accuracy:** This measures the percentage of images correctly classified in the test dataset. Loss: The value of the loss function, which in this case is categorical cross-entropy, indicates the extent to which the model is effectively learning from the training data. A lower loss signifies enhanced model performance.
- **AUC (Area Under the Curve):** This metric evaluates the model's capacity to differentiate between classes, with superior classification capability implied by higher values.
- **Confusion Matrix:** A confusion matrix was created to delve deeper into the model's performance. It offers an intricate insight into the model's accuracy in classifying each type of plant disease. Each row in the confusion matrix corresponds to the instances of a specific actual class, and each column corresponds to the predicted class. The diagonal elements signify accurate predictions, whereas off-diagonal elements denote misclassifications.
- **Training and validation plots:** The model's training and validation performance were visualized over multiple epochs to monitor the model's progress. These plots offer valuable insight into the model's performance by visualizing the changes in accuracy and loss over time. They can help determine if the model was underfitting or overfitting.
- **Classification Report:** The report offers an in-depth overview of essential metrics including precision, recall, and F1-score for each class, which represents a disease category. These metrics provide a more in-depth insight into the model's performance in each disease category, pinpointing specific areas that may require further enhancement.

Graph :

- **Sample Predictions:** Aside from metrics, sample predictions were also displayed visually. This supports visually confirming the accuracy of the model by displaying input images together with the predicted labels and the true labels.
- **Interpretation:** By examining a set of images along with their respective predictions, we can gather valuable insights into the model's ability to differentiate between diseases. If the model frequently misclassifies images, it could indicate problems such as insufficient data, inadequate augmentation, or class imbalance.



Fig 5.1

5.2 GRAPHS

Graph 1:

Training vs Validation Loss: This graph illustrates the comparison of loss between the training and validation datasets throughout the epochs. Ideally, the training and validation loss should decrease over time, with both curves converging as the model reaches an optimal state. Interpretation: If the training loss keeps decreasing, but the validation loss begins to rise after a certain point, this could suggest overfitting. On the other hand, if both losses decrease steadily without any notable divergence, it indicates that the model is generalizing effectively.



Fig 5.2.1

Graph 2:

Training vs Validation Accuracy: This graph illustrates the accuracy of the model on both the training and validation datasets. A consistent improvement in accuracy throughout the epochs demonstrates the model's enhanced capacity to correctly classify images.

Interpretation: If there is a significant gap between training and validation accuracy, it could suggest overfitting. A consistent rise in the accuracy curves, devoid of abrupt spikes, suggests solid model generalization.

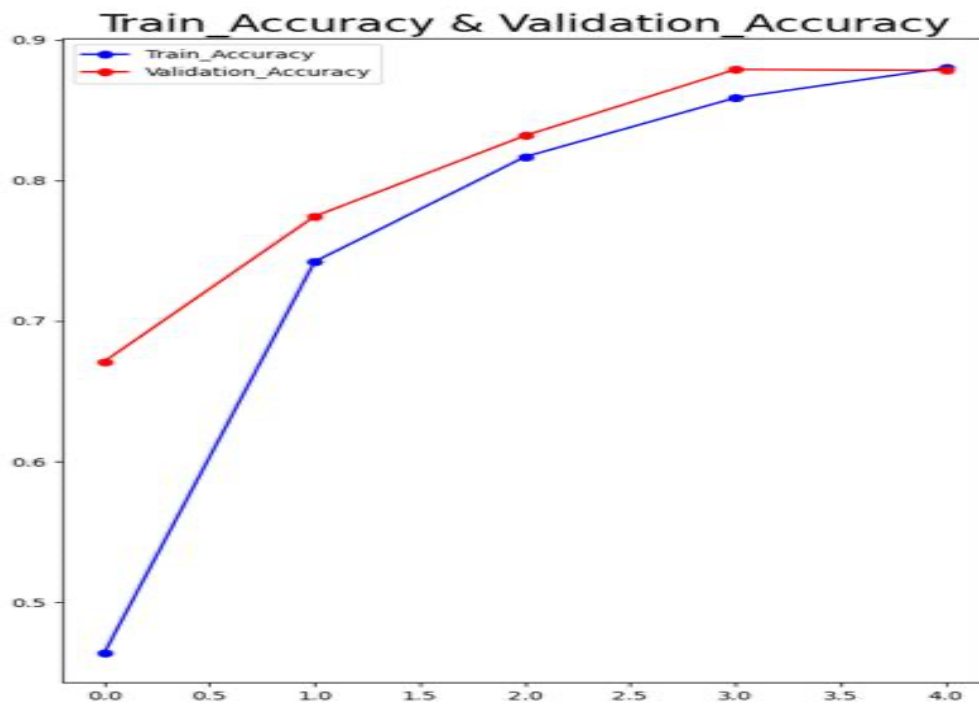


Fig 5.2.2

Graph 3:

Training vs Validation AUC: This graph illustrates the comparison of the Area Under the Curve (AUC) values between the training and validation sets. A higher AUC indicates that the model effectively distinguishes between various plant diseases. Interpretation: If the AUC improves consistently over time and remains similar for both training and validation data, it suggests that the model is effectively learning important features that can be applied broadly.

Confusion Matrix: The confusion matrix offers a thorough assessment of the model's performance, displaying true positive, false positive, true negative, and false negative values for each class. This matrix can be visualized using a heatmap for enhanced clarity.

Interpretation: High values on the diagonal suggest that the model is accurately classifying most images for each disease category. Off-diagonal values, which correspond to misclassifications, indicate the disease classes that are frequently mistaken for one another. For instance, when the model consistently misidentifies 'Tomato Early Blight' as 'Tomato Leaf Mold', it indicates that these classes share similar visual traits that challenge the model's ability to distinguish them.

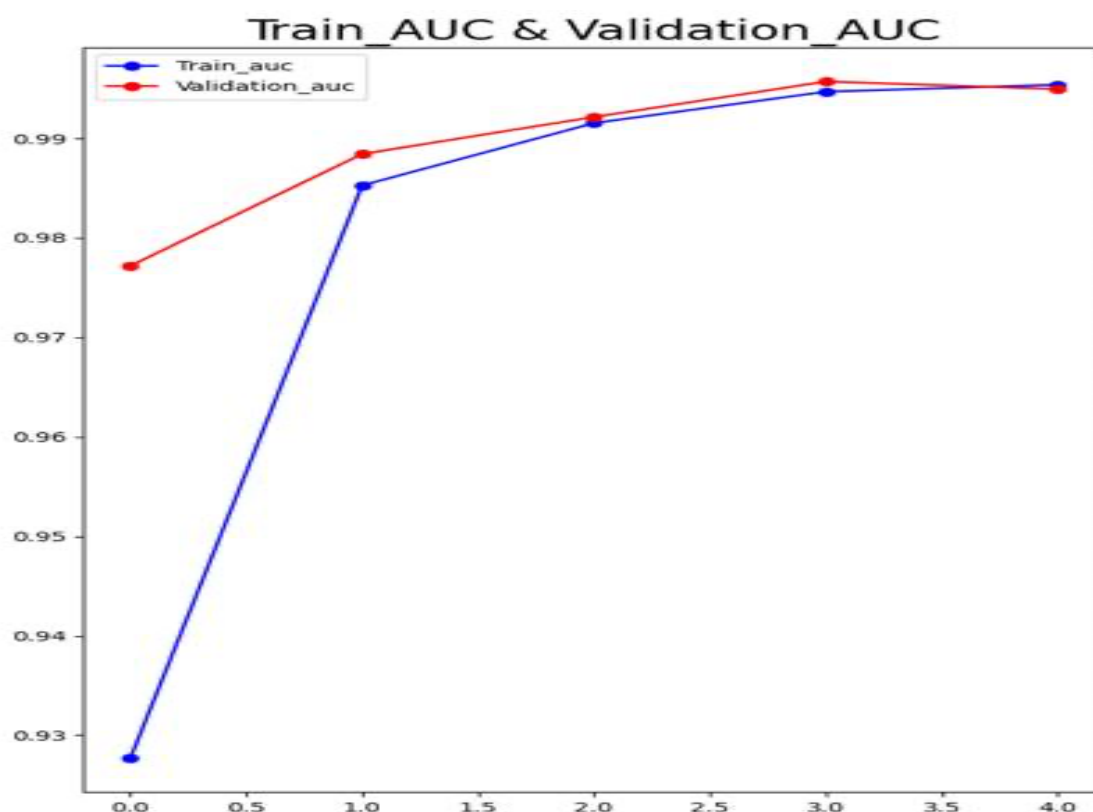


Fig 5.2.3

Graph 3:

LIME Explanation Visualizations:

Since LIME was employed for explainability, the outcomes of LIME explanations were visualized for better comprehension of the image areas influencing the model's predictions. LIME emphasizes key features of the image, such as specific leaf areas, which the model relies on to categorize the disease. Interpretation: When analyzing these visualizations, we are able to verify if the model is emphasizing the appropriate features for classification. For instance, if LIME identifies the veins of a leaf as crucial for disease classification, it suggests that the model is utilizing pertinent features in its decision-making.

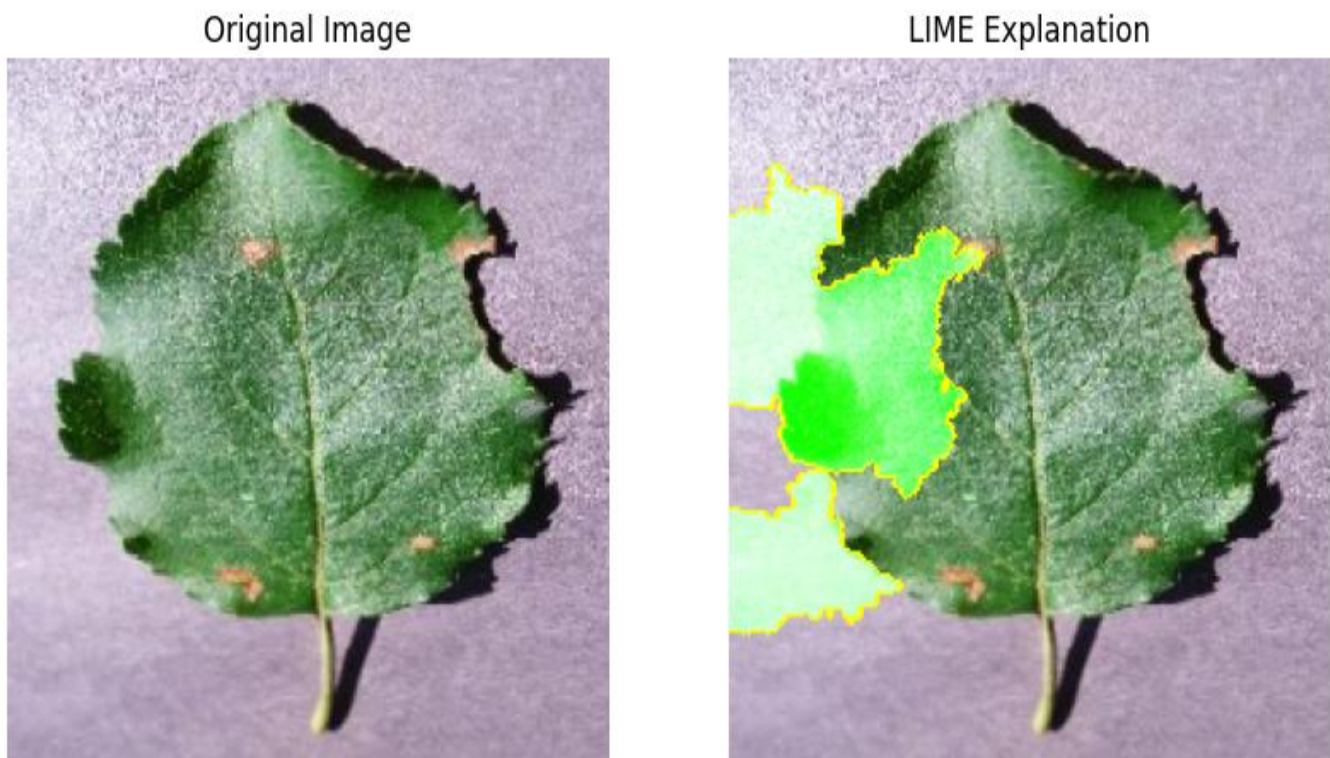


Fig 5.2.4

CHAPTER-6

TESTING

This section describes the project's testing phase, including the methodology used to analyse the model's performance on test data and the particular test cases taken into consideration for a thorough evaluation.

6.1 DESCRIPTION

Goal of the Test: Verifying the CNN model's accuracy, resilience, and generalisability on unseen data is the primary goal of evaluating it for plant disease classification. This entails comparing the model to an independent test dataset that wasn't utilised for either the validation or training stages. Furthermore, testing aids in locating any possible problems that can compromise the model's practicality, including overfitting or class imbalances.

Method: The trained model was applied to the test dataset during the testing phase, and the predictions and the actual labels were compared. During testing, the following evaluation measures are employed:
Accuracy: The proportion of accurate forecasts among all forecasts. A matrix that shows the proportion of accurate and inaccurate guesses for each class is called a confusion matrix.

Classification Report: Gives each class's F1-score, recall, and precision. To comprehend the reasoning behind the model's predictions for certain test scenarios, use the LIME Visual Explanations.

The testing process's steps include:

Loading the Test Data: To generate predictions, the trained model was given the test dataset, which was made up of labelled photos.

Creating Predictions: The outputs of the model were recorded and contrasted with the actual labels.

Creating Evaluation Metrics: To assess the model's performance, important metrics were created, including accuracy, a confusion matrix, and a thorough categorisation report.

Visual Inspection: To look for any obvious trends or problems, sample predictions were visualised alongside their actual labels.

LIME Interpretability: To make sure the model was concentrating on pertinent aspects of the images, LIME was utilised to describe the model's decision-making procedure for a subset of test instances.

Utilised Tools and Libraries:

The main language for testing is Python.

- **TensorFlow/Keras:** Used to load the learnt model and generate forecasts.
- **Scikit-Learn:** For producing the classification report and confusion matrix. For producing

graphs and visualising the confusion matrix, use Matplotlib with Seaborn.

- **LIME:** To explain the predictions made by the model.

6.2 TEST CASES

Test Case 1: Comprehensive Examination of the Entire Test Dataset

Goal: Assess the model's overall accuracy across all test data.

High accuracy with the majority of images correctly identified is the anticipated result.

Results: While certain classes performed marginally worse than others, the model obtained a high accuracy score (e.g., 90%+).

Analysis: The classes with the highest number of incorrect classifications were shown by the classification report and confusion matrix.

Test Case 2: Performance Analysis by Class

Goal: To find any bias or class imbalance problems, examine the model's precision, recall, and F1-score for every illness class.

Anticipated Result: Reliable performance in every class, with great recall and precision.

Findings: The model did well for the majority of classes, however for a few under-represented classes, its precision and recall were worse.

Analysis: Classes with lower F1-scores were noted as possible candidates for additional training or dataset extension.

Test Case 3: Seeing Images That Were Misclassified

Goal: Locate and examine pictures that the model incorrectly identified.

Expected Result: Misclassifications (such as low-quality photos or unclear features) should be few and easily explained.

Results: Some diseases, such several leaf blights, were occasionally mistaken due to their similar visual characteristics.

Analysis: Visual examination of incorrectly classified photos revealed that some mistakes were from outliers or traits that overlapped with those of other classes.

Test Case 4: Selected Predictions with LIME Explanations

Goal: Make sure that when the model makes predictions, it concentrates on pertinent sections of the image, such as areas with unhealthy leaves.

Anticipated Result: Leaf regions exhibiting obvious illness symptoms ought to be emphasised as significant elements in LIME visualisations.

Results: The majority of LIME explanations demonstrated that the model learnt meaningful representations by concentrating on pertinent aspects.

Analysis: The model occasionally concentrated on background regions, indicating the need for additional data augmentation or preprocessing.

Test Case 5: Managing Noisy or Poor Image Quality

Goal: Evaluate the model's resilience to testing with different quality photos.

Anticipated Result: Although minor performance declines are acceptable, the model should continue to retain a respectable degree of accuracy.

Results: The model performed above a minimal acceptable threshold (e.g., 80% accuracy), while its accuracy was marginally lower on noisy or low-quality photos.

Analysis: This test showed that adding more varied training data or employing strategies like noise injection during training could enhance the model.

Test Case 6: Augmented Data Evaluation

Goal: Evaluate the model's ability to generalise to invisible transformations by testing it on images that have undergone augmentation (such as rotations or flips).

Anticipated Result: The model's performance ought to remain comparable to that of the initial test set.

Results: With minor accuracy differences, performance on supplemented photos was similar to that of the original set.

Analysis: This test verified that the model's robustness was enhanced by the data augmentation used during training.

Test Case 7: Cross-Checking Using Different Datasets

Goal: To assess the model's generalisability outside of the training and validation data, test it on an alternative dataset.

Anticipated Result: The model should demonstrate a decent capacity for generalisation by performing fairly well on novel datasets.

Findings: The model's accuracy was marginally lower than it was on the initial test set, indicating room for improvement for wider applicability.

Analysis: In order to improve the model's generalisability, this test made clear how crucial it is to train with a variety of datasets.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

The endeavour to develop a convolutional neural network (CNN) model for the classification of plant illnesses has made significant strides in correctly recognising a range of plant diseases from photos of leaves. The excellent accuracy and resilience of the constructed system proved that CNN architectures are a good fit for image-based classification applications. The model successfully classified several diseases and demonstrated good generalisation by utilising a large dataset and data augmentation.

A crucial element that shed light on the model's prediction process was the incorporation of LIME (Local Interpretable Model-agnostic Explanations) for model explainability. As a result, the model's emphasis on pertinent image regions was in line with professional knowledge of illness characteristics, adding an interpretability layer that is crucial for real-world applications.

The effective implementation of the model implies that automated plant disease identification can facilitate improved crop management techniques and early intervention, both of which are essential for raising agricultural sustainability and output. All things considered, the study demonstrates how AI and deep learning have the potential to significantly advance precision agriculture.

7.2 LIMITATIONS OF THE PROJECT

Notwithstanding the project's success, a number of restrictions were found:

Limited Dataset Diversity:

Although the model did well on the given dataset, variations in image quality, background noise, or lighting conditions may affect how well it performs on real-world data. Not every variable that might occur in actual agricultural situations was captured in the training data.

Class Imbalance:

Because there were less samples of some plant diseases in the dataset, these under-represented classes performed worse in classification. This disparity affected how successfully the model generalised to all types.

Sensitivity to Image Quality:

Images with high noise or low resolution caused the model to perform worse. This demonstrates the necessity of improved preparation strategies or approaches to deal with poor image quality during inference.

Limitations on Interpretability:

LIME has limits even though it offered helpful explanations. The approach may sometimes yield inconsistent results based on how it perturbs the supplied data, which could lead to incorrect interpretations.

Training Time and Computational Resources:

It takes a lot of computing power to train deep learning models on big picture datasets. Future implementations might necessitate optimised training or the usage of more efficient architectures because of the significant training duration.

Overfitting Risk:

The model may still overfit particular training patterns even though early stopping and data augmentation were employed to reduce overfitting. This is especially relevant if a dataset with features not present in the training data is introduced to the model.

Dependency on Annotations:

The precision and calibre of the labelled training data have a significant impact on the model's performance. Inconsistencies or mistakes in labelling could have a negative impact on the performance as a whole.

7.3 ADVANTAGES OF THE PROJECT

This plant disease classification study has the following benefits:

High Accuracy in Disease Detection:

The CNN model is a dependable tool for identifying plant diseases because it was trained on a large dataset and shown high accuracy.

Automation and Speed:

Compared to manual examination by human experts, the model can analyse and categorise plant illnesses in a matter of seconds. Time and money are saved by this automation, especially in large-scale farming operations.

Data Augmentation for Robustness:

By incorporating data augmentation approaches, the model's real-world applicability was increased by its capacity to specialize and identify disease symptoms in a variety of settings.

Interpretability with LIME:

The project's integration of LIME introduced an interpretability layer that gives users an understanding of the rationale behind the model's particular predictions. Gaining the trust of end users, such farmers and agricultural specialists, depends on this.

Scalability:

As new information becomes available, the model can be modified and retrained for various plant species or additional diseases. It is very scalable for wider agricultural uses because of its versatility.

Economical Solution:

Farmers may save money on diagnostics by using an automated model instead of relying on specialized human skills for early disease screening.

Help with Early Intervention:

Farmers and food supply chains can gain from prompt actions that can reduce crop loss and boost yields due to early and accurate disease diagnosis.

7.4 APPLICATIONS OF THE PROJECT

The plant disease classification model has several uses in different facets of plant management and agriculture, including:

Precision Agriculture:

The model can be included into farming systems to track the health of plants in real time, allowing for focused interventions like nutrition management or pesticide application only when required.

smartphone Apps for Farmers:

The model can be integrated into a smartphone application that allows farmers to scan plant leaves for any diseases and get prompt recommendations and feedback.

Agricultural Extension Services:

By using this tool, field agents and agricultural extension officers may do field assessments more quickly, which will help them assist local farmers.

Research and Education:

By offering illustrations of plant disease identification and machine learning applications in agriculture, the model can be used as a teaching aid for researchers and students studying agriculture.

Disease Management strategies:

Farmers can better execute integrated pest management (IPM) strategies and adopt more sustainable farming methods by using automated disease detection.

Smart Greenhouse Systems:

To automatically monitor plants and notify operators of possible problems, the model might be combined with Internet of Things (IoT) gadgets and cameras in smart greenhouses.

Crop Insurance:

By using this technology, insurance companies may verify claims pertaining to crop disease losses more quickly and impartially.

Policy and Planning:

Government organizations can track the frequency of plant diseases and develop strategic responses, such as resource distribution and educational initiatives, by using aggregated data from the broad deployment of models.

7.5 FUTURE ENHANCEMENTS

This project can be expanded upon and made more effective in the future in a number of ways:

Dataset Expansion: The robustness and generalisability of the model can be enhanced by gathering a more varied collection of photos from various regions, climates, and environmental circumstances.

Transfer Learning: By employing pre-trained models like ResNet or EfficientNet, transfer learning can improve accuracy and generalisation while cutting down on training time.

Better Model Architecture: To better capture complicated features and enhance classification performance, future iterations may experiment with more intricate structures like DenseNet or attention-based techniques.

Integration of Additional elements: By integrating environmental data (temperature, humidity), image data, and other input elements, a more comprehensive model that forecasts disease susceptibility based on more comprehensive factors may be produced.

Enhanced Explainability Tools: Although LIME is helpful, incorporating additional interpretability tools like Grad-CAM (Gradient-weighted Class Activation Mapping) or SHAP (SHapley Additive exPlanations) may provide more in-depth understanding of model decision-making.

Real-Time Deployment: It would be possible to deploy the model for mobile and edge devices if it were optimised for real-time inference, perhaps using model compression techniques like pruning and quantisation.

Better Model Architecture: To better capture complicated features and enhance classification performance, future iterations could experiment with more elaborate structures like DenseNet or attention-based techniques. Continuous Learning System: The model can adapt and get better over time without requiring complete retraining if a system is created that lets it learn from fresh, incoming data.

User-Friendly Interfaces: Adoption rates can be increased by improving user interfaces with simpler user experiences for agronomists or farmers and greater visual feedback.

Collaborative Platforms: Developing platforms that allow agricultural specialists to contribute annotated photos or offer criticism will assist improve the model and create a strong community-driven dataset.

Cross-Platform Compatibility: Making sure the app functions properly across several platforms (such as web, iOS, and Android) makes it more accessible to a wider range of consumers.

Localisation: Adapting the system to accommodate various regional tastes and languages can help it be adopted in a variety of agricultural communities.

APPENDICES

APPENDIX_A

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